

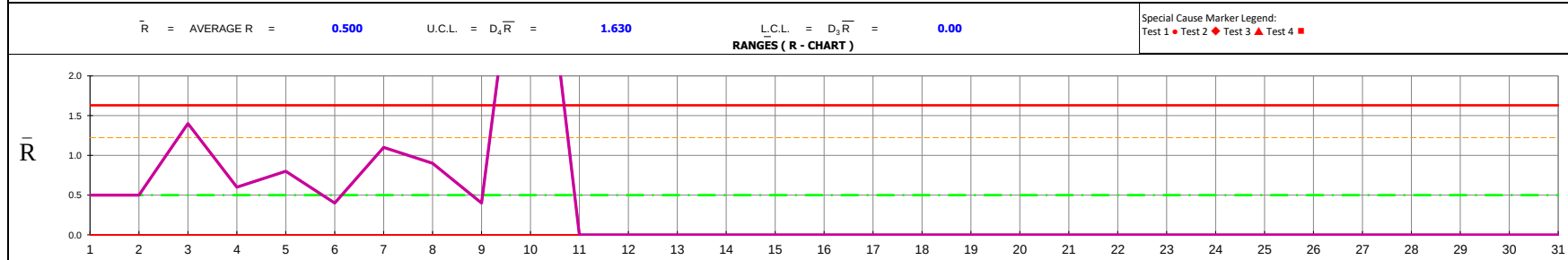
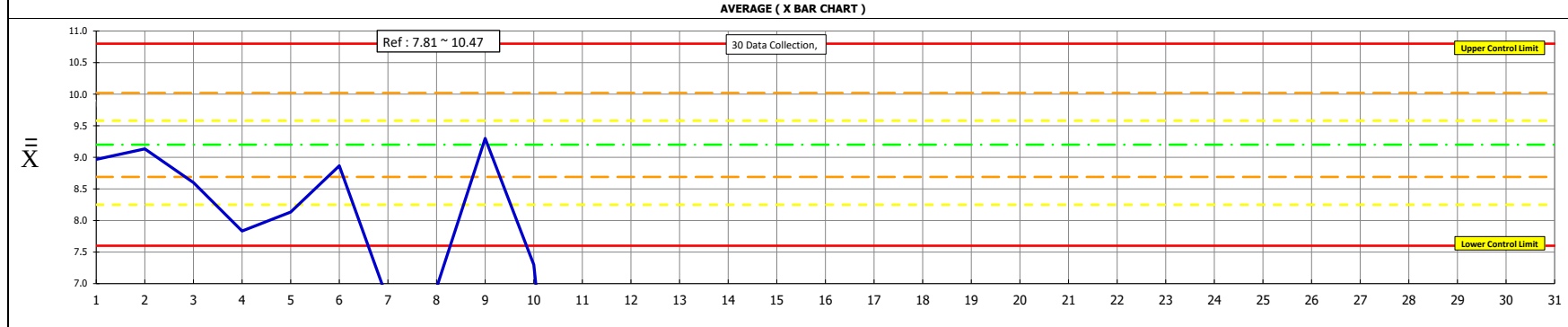


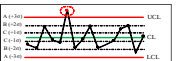
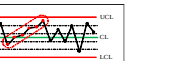
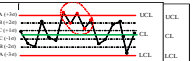
PT.SUMITOMO ELECTRIC WINTEC INDONESIA

 $\bar{X}$ - R CONTROL CHART

No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

MACHINE BDV Tester	SECTION QC	OPERATION Twister	DATE CONTROL LIMITS CALC.	SPECIFICATION min 3.7 max	COURSE NO.	CONTROL ITEM (Δ) YES NO
	MONTH July-25	CHARACTERISTIC Voltage		SAMPLE SIZE / FREQUENCY n = 3	TYPE / SIZE : D2IDW4-B-2 / 0.70 PIC : SUWARNO	
X = AVERAGE X = 9.200		U.C.L. = $\bar{X} + A_2 \bar{R}$ = 10.8000		L.C.L. = $\bar{X} - A_2 \bar{R}$ = 7.6000		



Special Cause	Titik berada di luar garis merah atau lebih dari 3 σ  6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun  4 atau 5 titik lebih dari 1 σ dengan posisi yang sama dan titik keluar dari garis merah atau 3σ 																															
	Month																															
Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	
DATA	1	9.200	8.900	7.900	7.600	8.000	8.700	6.900	6.500	9.300	10.600																					
	2	9.000	9.400	8.600	7.700	7.800	9.100	7.200	7.400	9.100	5.400																					
	3	8.700	9.100	9.300	8.200	8.600	8.800	6.100	6.900	9.500	5.900																					
N	3	3	3	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
AVERAGE, X	8.967	9.133	8.600	7.833	8.133	8.867	6.733	6.933	9.300	7.300	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	
Range	0.500	0.500	1.400	0.600	0.800	0.400	1.100	0.900	0.400	5.200																						
Moving Range		0.002	0.768	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000		
SHIFT 1																																
SHIFT 2																																

SPECIAL CAUSES
☒ TEST 1: ANY POINT OUT OF CONTROL LIMIT
☒ TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE
☒ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
☒ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
☒ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
☒ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
☒ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
☒ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION
1. _____
2. _____
3. _____
4. _____
5. _____

SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

CALCULATION
X = 8.1800
R = 1.1800
S _{xy} = 0.6970 S _{yy} = 1.223
THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED
C _{pk} = MIN { $\frac{\bar{X} - LSL}{3S_{xy}}$, $\frac{USL - \bar{X}}{3S_{yy}}$ } = MIN { 0.28 , 1.25 } = 0.28
C _p = $\frac{USL - LSL}{6S_{yy}}$ = { 0.77 }

CHECKED BY	APPROVED BY
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